

RentalHub

INTRODUCTION:

RENTALHUB – EQUIPMENT RENTAL MANAGEMENT WEB APPLICATION:

RentalHub is a web-based equipment rental management application designed to provide a simple and efficient platform for users to rent equipment online. The application allows users to browse available equipment, check its availability, make rental bookings, manage their rentals, and provide reviews. The system is developed using modern web technologies such as React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for database management. These technologies work together to provide a responsive, user-friendly, and reliable rental platform. RentalHub helps users avoid the difficulties of traditional equipment rental processes by providing important rental information in one place. Users can view equipment details, check availability, select rental dates, and manage their bookings through the application. The system also provides management features for handling equipment, users, bookings, rental and return information, and reviews. By connecting the frontend, backend, and MongoDB database, RentalHub provides an end-to-end equipment rental management solution. The main objective of RentalHub is to make the equipment rental process simple, organized, accessible, and efficient for both users and administrators.

SCENARIO BASED CASE STUDY:

Consider a user who needs to rent equipment for a short period of time. Instead of visiting multiple rental shops and checking the availability of equipment manually, the user can use the RentalHub web application.

The user first creates an account and logs into the RentalHub system. After logging in, the user can browse the available equipment and view important details such as equipment name, category, description, rental price, and availability.

If the required equipment is available, the user can select the required rental dates and place a booking through the application. The booking information is stored in the MongoDB database and can be accessed through the user's account.

The user can also view their current and previous rental bookings. After using the equipment, the rental can be marked as returned, allowing the system to maintain updated rental and availability information.

After completing a rental, the user can provide a review for the equipment. This helps maintain feedback information within the system.

User Workflow

1. Registration & Login

The user creates an account and logs into RentalHub.

2. Browse Equipment

The user views available equipment and checks its details.

3. Check Availability

The user selects the required rental dates and checks whether the equipment is available.

4. Booking

The user books the selected equipment for the required period.

5. Rental Management

The user can view and manage their active and previous rentals.

6. Return Equipment

After completing the rental period, the equipment is marked as returned and its availability is updated.

7. Review

The user can provide a review based on their rental experience.

Expected Outcome

RentalHub provides a complete online equipment rental workflow where users can **register, browse equipment, check availability, book equipment, manage rentals, return equipment, and provide reviews** through a single web application.

SYSTEM REQUIREMENTS:

1. SOFTWARE REQUIREMENTS:

The following software and technologies are required to develop and run the RentalHub application:

Software / Technology Purpose

Operating System	Windows 10/11 or any compatible operating system
Visual Studio Code	Used for writing and managing the project code
React.js	Used to develop the frontend user interface
Node.js	Used to run the backend JavaScript environment
Express.js	Used to create backend APIs and server-side functionality
MongoDB	Used to store and manage application data
MongoDB Compass	Used to view and manage MongoDB databases
Git & GitHub	Used for version control and source-code management
Web Browser	Used to run and test the RentalHub application
npm	Used to install and manage project dependencies

2. HARDWARE REQUIREMENTS:

The following hardware configuration is recommended for developing and running the RentalHub application:

Hardware	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB minimum, 8 GB recommended
Storage	At least 10 GB of free space
Display	1366 × 768 resolution or higher
Internet	Stable internet connection
Keyboard & Mouse	Standard keyboard and mouse

3.TECHNOLOGY STACK:

The RentalHub application uses the following technology stack:

- **Frontend:** React.js
- **Backend:** Node.js + Express.js
- **Database:** MongoDB
- **Development Tool:** Visual Studio Code
- **Version Control:** Git and GitHub
- **API:** REST API

PROJECT ARCHITECTURE:

RentalHub follows a modern, scalable three-layer web application architecture consisting of the **Frontend Layer**, **Backend Layer**, and **Database Layer**.

Architecture Flow Diagram



Frontend Layer:

The frontend of RentalHub is developed using **React.js**. It provides the user interface through which users can register, log in, browse equipment, check availability, make bookings, manage rentals, and submit reviews.

Backend Layer:

The backend is developed using **Node.js and Express.js**. It handles business logic, user authentication, equipment management, availability checking, booking operations, rental/return management, and review operations.

The backend communicates with the frontend through **REST APIs**.

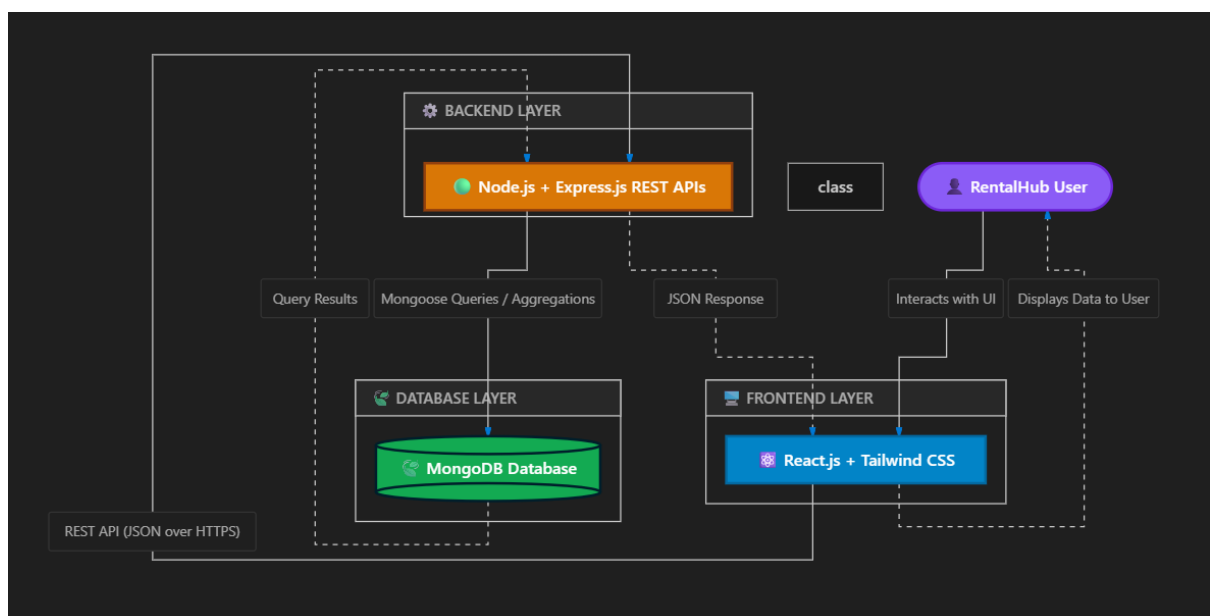
Database Layer:

MongoDB is used as the database for RentalHub. It stores information such as user details, equipment details, bookings, rentals, returns, and reviews.

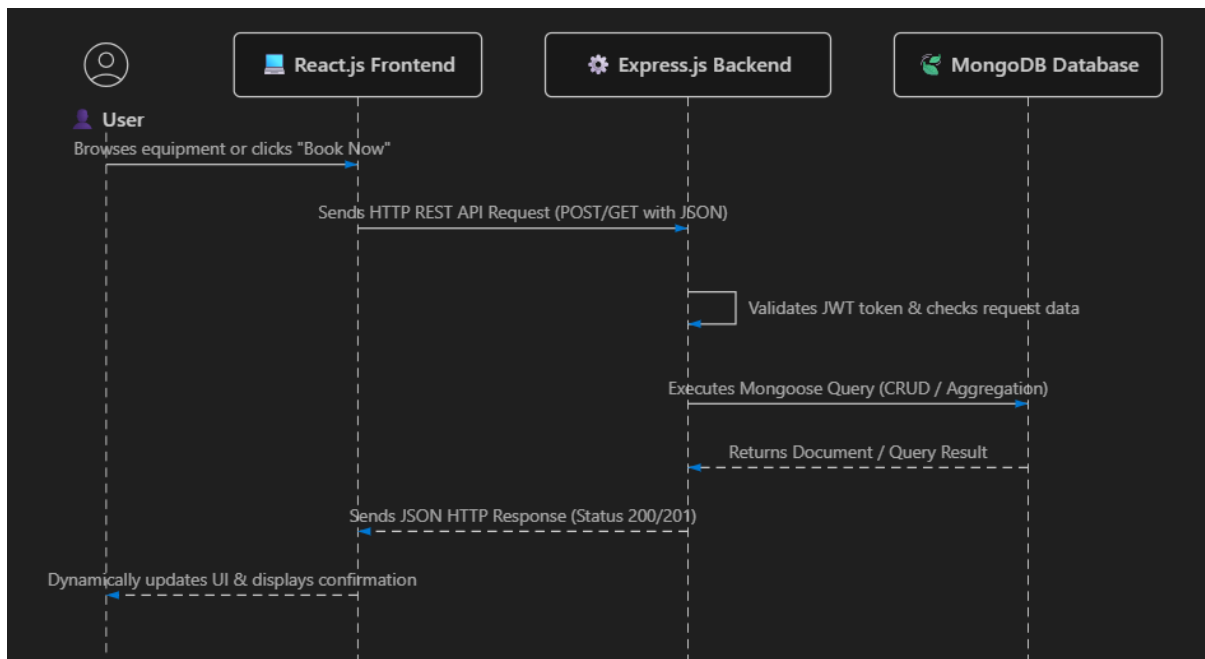
Data Flow:

1. The user interacts with the RentalHub React.js frontend.
2. The frontend sends a request to the Express.js backend through a REST API.
3. The backend processes the request and communicates with MongoDB.
4. MongoDB returns the required data or performs the requested operation.
5. The backend sends the response back to the React.js frontend.
6. The frontend displays the result to the user.

Alternative: Compact 3-Tier Diagram



Data Flow:



KEY FEATURES:

User Registration and Login:

Users can create an account and securely log in to the RentalHub application. User details are stored in the MongoDB database.

Equipment Browsing:

Users can browse the available equipment and view details such as equipment name, category, description, rental price, and availability.

Equipment Availability:

Users can check whether a particular equipment is available for their required rental period before making a booking.

Equipment Booking:

Users can select the required equipment and rental dates and place a booking through the RentalHub application.

Rental and Return Management:

Users can view their rental information and manage the return of rented equipment. The equipment availability is updated after the return.

Booking History:

Users can view their current and previous bookings through their account.

Reviews and Ratings:

After completing a rental, users can provide reviews and ratings for the rented equipment.

Admin Management:

The administrator can manage users, equipment, bookings, rentals, returns, and reviews through the system.

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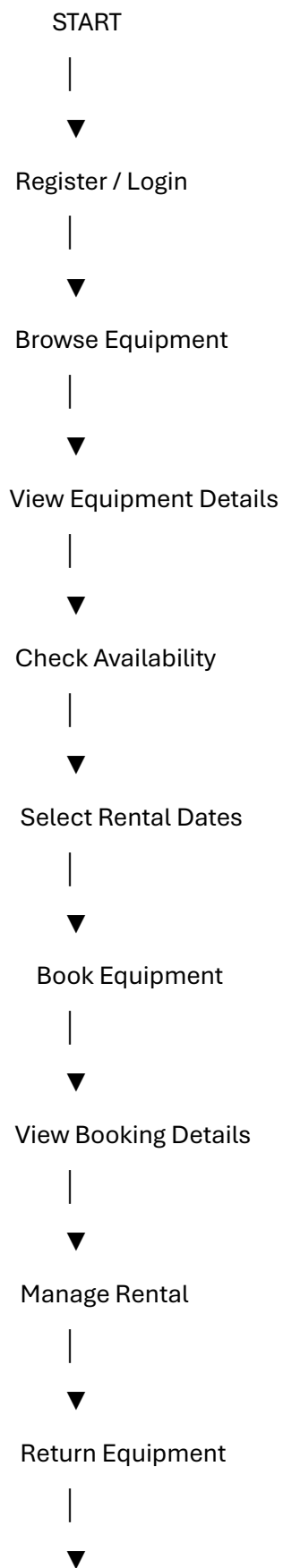
Main Workflow:

Register/Login → Browse Equipment → Check Availability → Book → Rent → Return → Review

7. USER FLOW

The RentalHub system provides a simple flow for users to complete the equipment rental process.

User Flow



Give Review



END

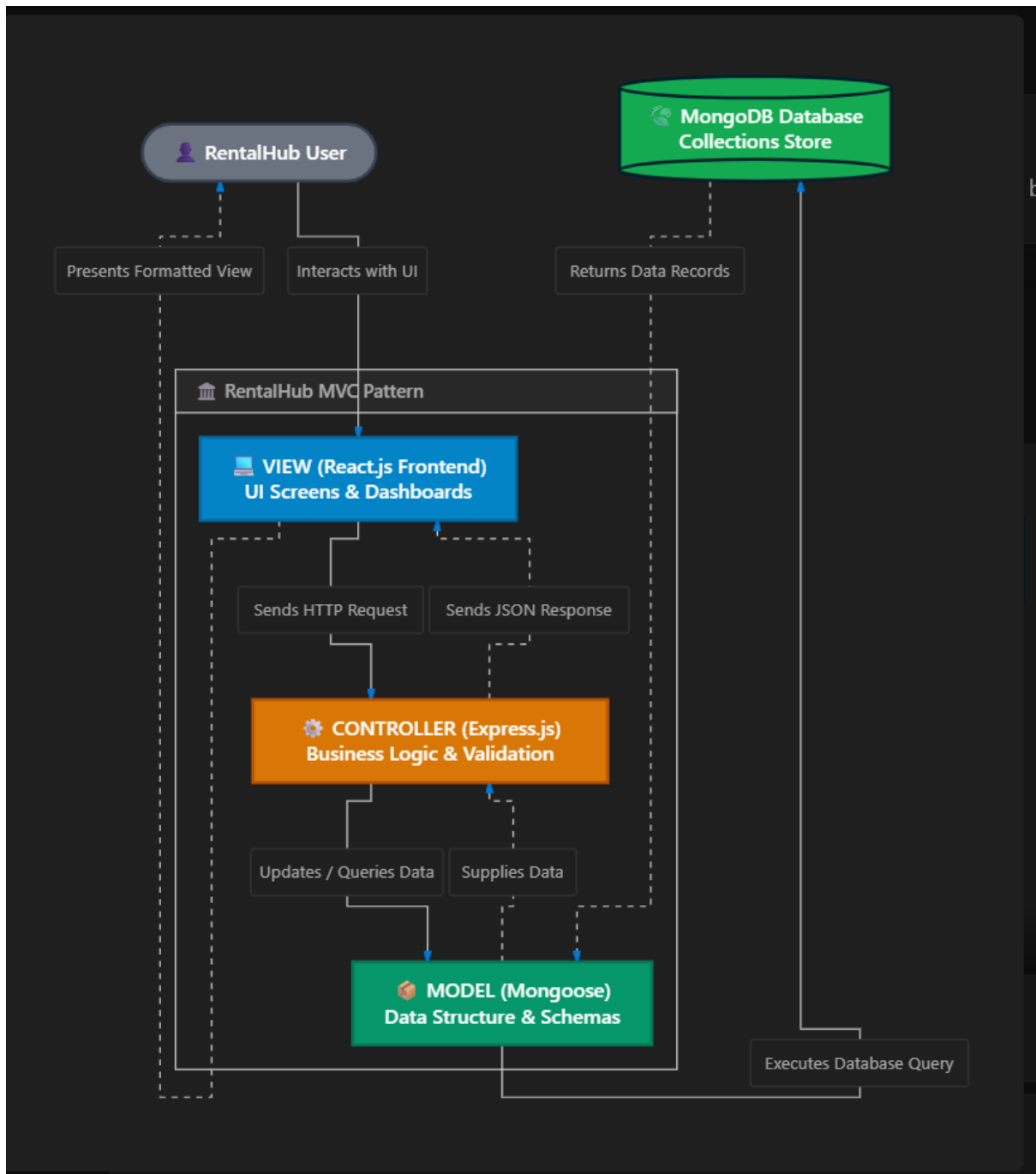
User Process

1. **Register / Login** – The user creates an account or logs into the existing account.
2. **Browse Equipment** – The user views the available equipment.
3. **View Details** – The user checks equipment information such as name, category, price, and description.
4. **Check Availability** – The user checks whether the equipment is available for the required rental period.
5. **Select Rental Dates** – The user selects the required rental start and end dates.
6. **Book Equipment** – The user confirms the booking.
7. **View Booking** – The user can view the booking details through their account.
8. **Manage Rental** – The user can track and manage the rental.
9. **Return Equipment** – The user completes the rental by returning the equipment.
10. **Give Review** – The user can submit a review and rating after completing the rental.

MVC PATTERN:

RentalHub follows the **MVC (Model–View–Controller)** architectural pattern to organize the backend application into separate components. This makes the application easier to develop, maintain, and manage.

MVC Architecture:



MODEL LAYER:

The **Model Layer** is responsible for managing the data and database structure of the RentalHub application. It defines how the application data is stored, retrieved, and managed in the MongoDB database.

RentalHub uses MongoDB as its database, and the Model Layer communicates with MongoDB to perform database operations.

Main Models in RentalHub

1. User Model

Stores user information such as name, email, password, and user role.

2. Equipment Model

Stores equipment information such as equipment name, category, description, rental price, and availability status.

3. Booking Model

Stores booking information such as user, equipment, rental start date, rental end date, and booking status.

4. Rental Model

Stores rental and return information, including rental details and return status.

5. Review Model

Stores user reviews and ratings for rented equipment.

Model Layer Flow

React.js Frontend

↓

Express.js Controller

↓

Model Layer

↓

MongoDB Database

The Model Layer helps maintain a structured and organized database system and provides the required data to the Controller Layer.

PROJECT SETUP AND CONFIGURATION:

The RentalHub project is divided into two main parts: **Frontend** and **Backend**. The frontend is developed using React.js, while the backend is developed using Node.js and Express.js. MongoDB is used as the database.

PROJECT FOLDER STRUCTURE:

The main project folder is created with separate frontend and backend folders.

RentalHub/

|

|— frontend/

|

| └─ React.js Application

|

```
└─ backend/
    └─ models/
    └─ controllers/
    └─ routes/
    └─ middleware/
```

```
└─ server.js
```

The React.js frontend is created using Vite.

```
npm create vite@latest frontend
```

The required options are selected during the setup, and React.js is selected as the framework.

After creating the frontend, the project dependencies are installed:

```
cd frontend
```

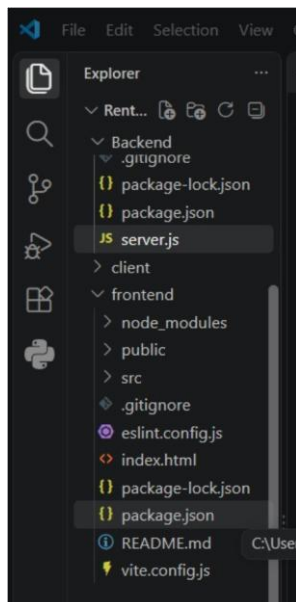
```
npm install
```

The frontend application can be started using:

```
npm run dev
```

The React development server provides the user interface for the RentalHub application.

FRONTEND STRUCTURE:



Frontend Reference video:

<https://drive.google.com/file/d/1qbNw177foB4wG8wDFhFJyEFBLUSoo28X/view?usp=drivesdk>

BACKEND SETUP:

A separate backend folder is created for the Node.js and Express.js application.

```
mkdir backend
```

```
cd backend
```

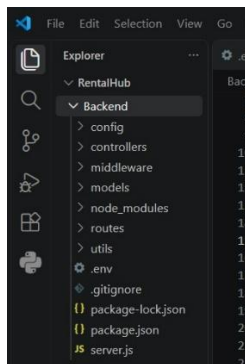
```
npm init -y
```

Express.js and other required dependencies are installed using npm.

```
npm install express mongoose cors dotenv
```

The backend server is configured using Node.js and Express.js

BACKEND STRUCTURE:



Backend Reference video:

<https://drive.google.com/file/d/1oXE1RF0do1iVYFlkimUu59NH64Bl4vz-/view?usp=drivesdk>

DATABASE CONFIGURATION:

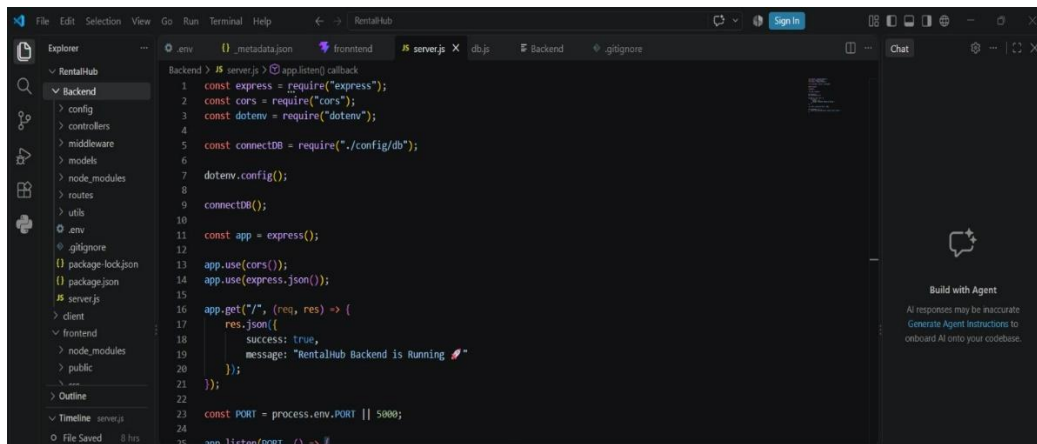
MongoDB is used to store RentalHub application data.

The backend connects to MongoDB using **Mongoose**. The database connection details are stored securely using environment variables.

Example:

```
MONGODB_URI=your_mongodb_connection_string
```

```
PORT=5000
```



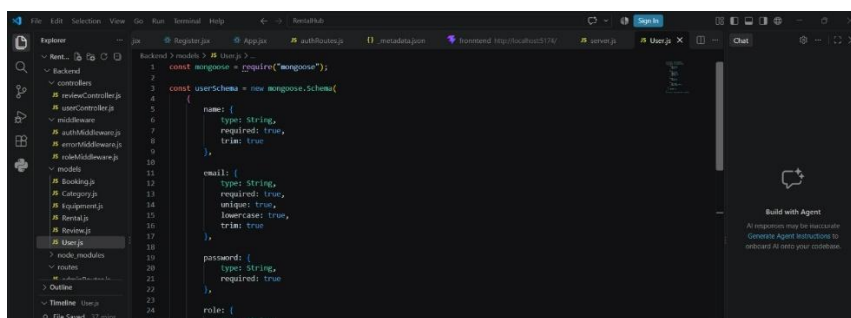
DEVELOPMENT ENVIRONMENT:

The RentalHub application is developed and tested using:

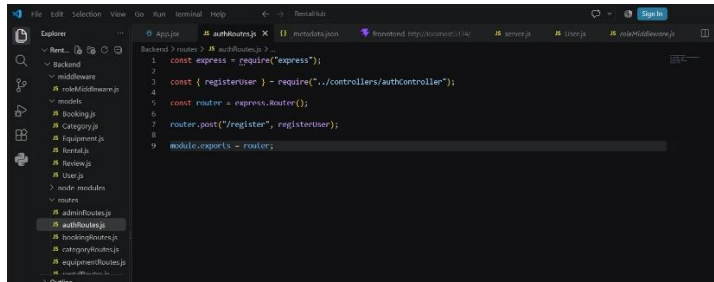
- **Visual Studio Code** – Code editor
- **React.js + Vite** – Frontend development
- **Node.js + Express.js** – Backend development
- **MongoDB** – Database
- **Mongoose** – MongoDB connection and data modelling
- **Git & GitHub** – Version control

- 1) **Users.js** - The user.js file defines the structure of user data in the RentalHub application. It is created inside the models folder and uses **Mongoose** to define the MongoDB schema.

The User Model stores basic user information required for registration, login, and role-based access.

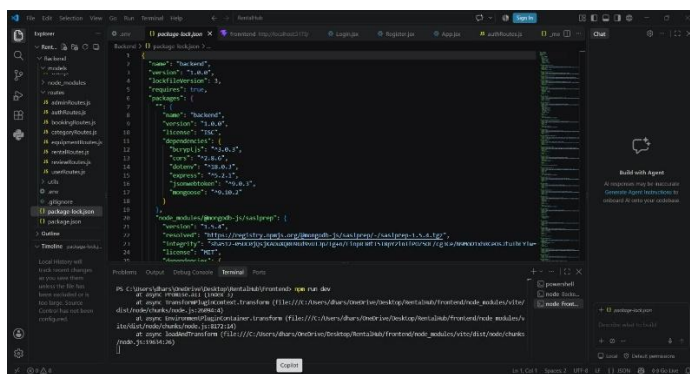


- 2) **AuthRoutes.js** - The AuthRoutes.js file defines the authentication-related API routes of the RentalHub application. It handles requests related to user registration and login.

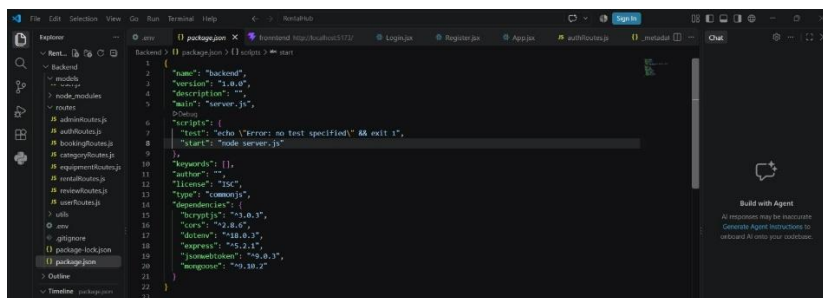


3) **package-lock.json** - The package-lock.json file is automatically generated by npm when dependencies are installed in the RentalHub project.

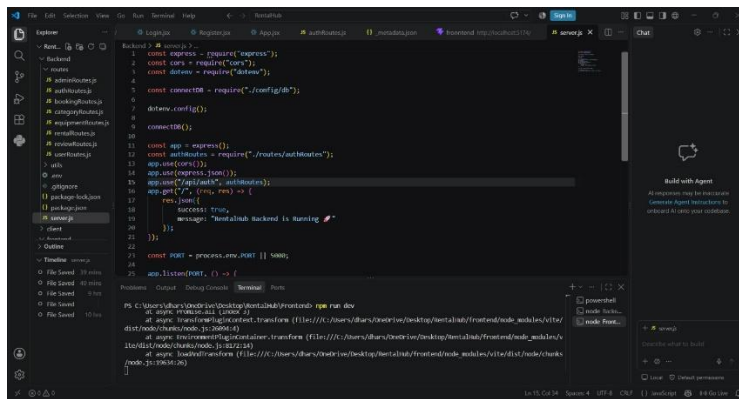
It records the exact versions of the installed packages and their dependencies. This helps ensure that the same package versions are installed when the project is run on another system.



4) **package.json** - The package.json file is the main configuration file for the Node.js backend project. It contains important information about the project, available npm scripts, and the packages required to run the application.



5) **server.js** - The server.js file is the main entry point of the RentalHub backend application. It is responsible for creating the Express server, configuring middleware, connecting to the MongoDB database, and registering API routes.



SRC:

The src folder is the main source directory of the React.js application. It contains the application pages, components, styles, and other frontend files.

The main files include:

- App.jsx – Main React application component.
- main.jsx – Entry point of the React application.
- index.css – Global CSS styles.
- user/ – Contains user-related pages.
- components/ – Contains reusable UI components.

USER:

- The user folder contains pages related to user activities in RentalHub.
- **Main User Pages**
- **Login.jsx**
Provides the login interface for existing users.
- **Register.jsx**
Allows new users to create a RentalHub account.
- **UserDashboard.jsx**
Displays user-related information such as bookings and rental details.

COMPONENTS:

The components folder contains reusable React.js components used throughout the RentalHub frontend application. These components help maintain a consistent user interface and reduce code duplication.

Components Structure:

frontend/

└─ src/

└─ components/

├─ Navbar.jsx

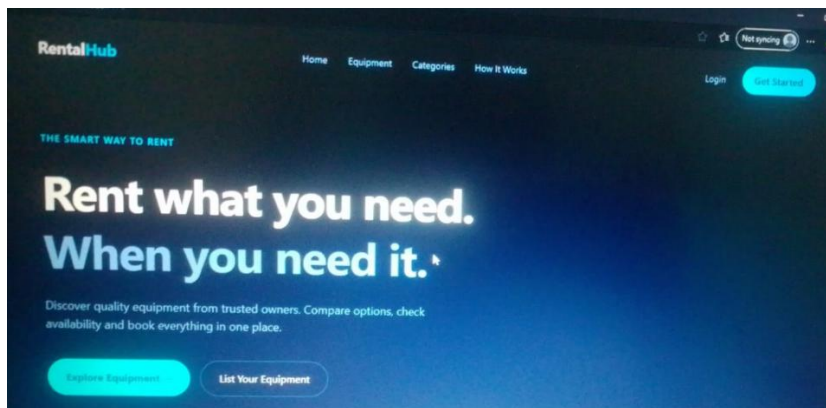
├─ EquipmentCard.jsx

├─ BookingForm.jsx

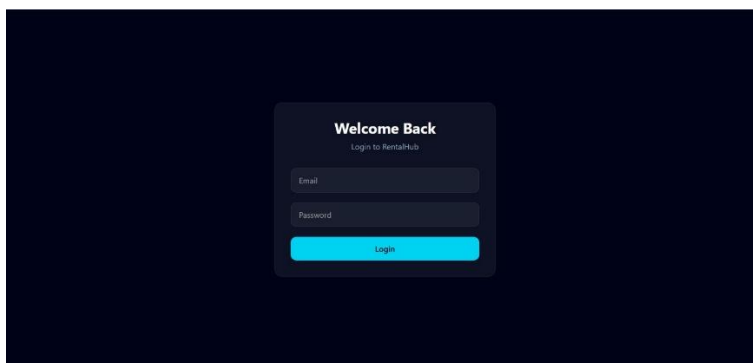
└─ Footer.jsx

Output ScreenShots:

Landing Page:



Login.jsx:



Register.jsx:

Create Account

Join RentalHub

Full Name

Email

Password

Register

Order Page:

RENTALHUB

Search Products...

Ravichandran

Admin

Logout

Dashboard

Products

Rentals

Active

History

Customers

Revenue Reports

Settings

RentalHub - Dashboard

Total Revenue

\$14,580.50

Active Rentals

48

Pending Returns

12

Rental Activity This Week

Current Rentals Overview

View All

Order ID	Product Name	Customer Name	Rent Date	Return Date	Price	Delivery Status	Payment
#RH1001	Professional Camera Canon R5	Arul Kumar	24 Oct 2023	30 Oct 2023	₹4,500	Delivered	Paid
#RH1002	Laptop, Laptop	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Delivered	Paid
#RH1003	DSLR, DSLR	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Out for Delivery	Paid
#RH1004	Projector, Projector	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Delivered	Paid
#RH1005	Tent, Tent	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Delivered	Paid
#RH1006	Car, Factor	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Out for Delivery	Paid
#RH1007	Drill, Trill	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Out for Delivery	Paid
#RH1008	Chair, Chair	Korntne	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Pending	Paid
#RH1009	Suit, Suit	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Pending	Paid
#RH1010	Bike, Stoker	Arul Kumar	24 Oct 2023	30 Oct 2023	₹1,500 - ₹7,000	Pending	Paid

Rents

Recent Activity Log

Ravichandran

Demo video link:

<https://drive.google.com/file/d/1HxqmtJIhhik2zWRIaWEQLRS6FPw6K40t/view?usp=drivesdk>